

ZHONGWEI (ZACHARY) ZHANG

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PERSONAL STATEMENT

Master of Information Technology candidate with professional experience in C++ backend development, API debugging, database optimisation and infrastructure monitoring. Interested in software engineering, full-stack web development and database systems. Brings hands-on experience building and deploying team-based applications, together with a practical and detail-focused approach to problem-solving.

KEY COMPETENCIES

- Diagnoses technical issues through structured debugging, log analysis, API testing and post-change verification.
- Collaborates effectively in four- to six-person project teams using Git, GitHub, branch-based development and clear task ownership.
- Takes ownership of assigned components and supports projects through implementation, integration, testing and deployment.
- Communicates technical findings and works with developers to resolve cross-layer application and database issues.
- Adapts quickly to unfamiliar technologies and independently completes complex problem-solving tasks.

TECHNICAL TOOLKIT

Languages: C++, TypeScript, JavaScript, SQL, Java, HTML, CSS

Frontend: React, Next.js, SvelteKit, Tailwind CSS, Zustand

Backend and APIs: Node.js, Express, REST APIs

Databases: PostgreSQL, MySQL, SQLite, Turso/libSQL

Testing and Delivery: Vitest, Git, GitHub, Vercel, Visual Studio

PROFESSIONAL EXPERIENCE

System Engineer, Tencent (China)

Oct 2024 - Apr 2025

C++ | MySQL | REST APIs | Internal Monitoring and Operations Tools

- Monitored server and infrastructure performance using internal operations tools, identifying abnormal service behaviour and issues affecting response times.
- Debugged REST APIs by sending test requests, examining parameters and responses, reviewing service logs, and comparing performance before and after changes.
- Analysed and modified server-side C++ code to resolve service-logic defects and address performance and memory-related issues.
- Collaborated with developers to investigate abnormal MySQL records and optimise table structures, indexes and slow queries, improving database query performance.
- Performed functional testing and post-change verification to confirm fixes and system stability.

PROFESSIONAL EXPERIENCE (CONTINUED)

Software Developer , Shanxi Xinhui Technology Co., Ltd. (China) <i>C++ MySQL Visual Studio Proprietary Version Control System</i> • Maintained and optimised C++ backend services for a customer-facing data management and analytics system. • Diagnosed and resolved software defects, crashes and memory-related issues by analysing existing code and troubleshooting backend components. • Improved performance by reducing redundant processing and optimising loops, data structures and memory usage. • Wrote SQL queries, maintained application data and modified MySQL table structures; used the company's proprietary version control system for code submissions and version tracking.	Sep 2021 - Jul 2024
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SELECTED SOFTWARE PROJECTS

Last Answer - Browser-Based Fantasy Quiz RPG Six-Person Team GitHub Live Demo <i>Next.js React TypeScript Tailwind CSS Zustand PostgreSQL Vitest Vercel</i> • Developed the quiz-battle subsystem, implementing deterministic TypeScript combat logic and Zustand-based state management for timed questions, combo tracking, support tools, boss encounters and reward transitions. • Created four tactical support mechanics and a performance-based reward system driven by answer accuracy, response speed and combo state. • Wrote Vitest tests covering combat calculations, state transitions, timing and combo rules, reward logic and component interactions. • Integrated battle features with backend APIs and supported production release by connecting the backend to PostgreSQL and validating the deployed version.	2026
DoraBlog - Full-Stack Blogging Platform Four-Person Team GitHub Live Demo <i>SvelteKit Node.js Express SQLite Turso/libSQL Vercel Blob REST APIs</i> • Developed end-to-end authentication and role-based authorisation using SvelteKit and Express, including bcrypt hashing, JWT generation, HttpOnly cookies, authentication middleware, token refresh, secure logout and administrator-only API protection. • Implemented profile management, avatar uploads, account deletion, and article search and sorting through frontend and REST API development. • Integrated the web backend with an existing Java administration client and resolved API, session and database issues across system layers. • Independently deployed the application to Vercel, migrating SQLite data to Turso/libSQL and filesystem-based uploads to Vercel Blob for persistent serverless hosting.	2025 - 2026

EDUCATION

Master of Information Technology , University of Auckland (New Zealand)	Jul 2025 - Present
Bachelor of Mechanical Engineering , Xi'an Jiaotong University (China)	Sep 2017 - Jul 2021